Uniform mesh refinement underestimates the frequencies of the DeviceLayout.jl transmon: an independent finite-element cross-check and re-optimization

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Abstract

The SingleTransmon example of DeviceLayout.jl publishes the qubit and resonator frequencies of an initial and an optimized transmon design, computed with Palace. We reproduce them with Palace to within 0.7% and cross-check Palace with geode-fem, an independently implemented finite-element solver; on shared meshes the two agree to 10^{-4}% on the qubit and to 0.03% on the resonator. Under refinement the qubit frequencies keep rising. On the initial design the qubit of an adaptive Palace run reaches 4.4297 GHz, 7.0% above the published 4.14 GHz, with no extrapolation. Extrapolated at order 1 in the mesh size, the fastest rate the sheet-edge singularity allows and so an estimate that leans low, the two qubits lie 7–9% and the resonators 2.7% above their published values, and a certified bracket on the island capacitance, read through a circuit model, is consistent with that shift. Re-optimized for the 3.0/4.0 GHz targets at the extrapolated frequencies, the design needs a capacitor about 19–20% longer than published (about +17% to +39% within the extrapolation band). Newton's method in DeviceLayout's variables meets its stopping criterion in 3 Palace runs, with geode-fem's shape derivative or a scaling law in the Jacobian, where DeviceLayout's PRIMA optimizer takes 10 in the same environment.

1 Introduction

Superconducting-qubit layouts are designed by iterating field solves against frequency targets, a loop that QDesignOptimizer (Eriksson et al., 2025) steers with analytic models. DeviceLayout.jl (AWS Center for Quantum Computing, 2025a) builds the layout and its mesh, and Palace (Palace developers, AWS Center for Quantum Computing, 2023), an MFEM-based solver (Anderson et al., 2021), computes the eigenmodes. DeviceLayout's SingleTransmon example (AWS Center for Quantum Computing, 2025b; Peairs and Carson, 2025) runs the loop end to end. It publishes an initial transmon-plus-resonator design at 4.14 and 5.591 GHz and a design optimized for targets of 3.0 and 4.0 GHz, at 3.005 and 3.982 GHz, found with a derivative-free optimizer in nine Palace runs.

Those frequencies come from one discretization, and the example itself warns that they are sensitive to it. This paper measures how sensitive. Our main result is that the published frequencies are discretization-limited. Palace's adaptive refinement of the initial design reaches a qubit 7.0% above the published value without extrapolation, and its uniform ladders extrapolate to qubits 7–9% above both published values. geode-fem reproduces every level of those ladders it can reach to 10^{-4}% on the qubit, and a certified bound on the island capacitance, read through a circuit model, is consistent with that shift and caps the initial qubit under that model.

The second code is GEODE-FEM ("geode-fem"), a Rust finite-element solver on the Burn tensor framework (Tracel AI, 2024). It solves the same $H(\mathrm{curl})$ eigenproblem as Palace with a

separate implementation, and it computes design derivatives, among them the shape derivative of an eigenfrequency. On a mesh of DeviceLayout's initial design, given as the same input file to both codes, geode-fem reproduces Palace's six eigenmodes above 4.5 GHz to $\leq 0.033\%$ and the qubit mode to $5 \times 10^{-5}\,\%$ (Section 4). This agreement establishes that the two codes assemble and solve the same discrete problem. It also gives an independent same-mesh cross-check of the Palace/DeviceLayout.jl transmon stack, complementing the earlier checks of Palace by SQDMetal (Sommers et al., 2025) and Ye et al. (2025).

Our contributions are:

- **A refinement study of a published design example.** Resolution ladders at first and second order put the extrapolated qubits 7.3% and 8.7% above the published values and the resonators 2.7% above; a second refinement family and an adaptive run check the qubit estimate (Section 5).
- **Certified capacitance brackets with the sapphire tensor.** An equilibrated-flux lower bound and a conforming upper bound bracket the island capacitance of the unmodified device in $[79.56, 79.76]$ fF; under a two-node circuit model they cap the initial qubit at 4.477 GHz (Section 6).
- **A re-optimized design, with controls.** At the extrapolated frequencies the 3.0/4.0 GHz design needs $c \approx 1451$–$1466\,\mu$m against the published $1217.678\,\mu$m. Newton's method meets DeviceLayout's criterion in 3 Palace runs with the shape derivative or a scaling law in its Jacobian, the latter also with no geode-fem input at all; Broyden's method takes 5 and PRIMA 10 in the same environment (Section 7).
- **Same-input-mesh agreement with Palace** on six eigenmodes and the qubit mode, with the 17.49 GHz mode that both codes share traced to the distributed port model (Section 4).

Section 10 sorts the results by the evidence behind them and explains why the shape derivative gives no advantage on this benchmark.

2 Related Work

Cross-checks of Palace. SQDMetal (Sommers et al., 2025) cross-checks a Palace workflow against COMSOL and Ansys with mesh-convergence studies, and Ye et al. (2025) compare a Palace workflow with cryogenic measurement at the $\sim$0.3% level. Both use models built independently for each tool. Our comparison instead holds the input mesh and the junction model fixed and changes the solver implementation, and it adds a refinement study of a published design. In V&V terms (Oberkampf and Roy, 2010; Roache, 1997) it is code verification and code-to-code comparison, not validation against measurement. As of October 2026, neither Palace nor DeviceLayout.jl has a methods publication; we cite their repositories and documentation.

Design sensitivities and qubit-design optimization. Adjoint sensitivity analysis of frequency-domain electromagnetic solvers is an established field (Nikolova et al., 2004), including shape adjoints for Maxwell cavity eigenvalues (Akçelik et al., 2005); Palace exposes no derivatives. In qubit design, the SQcircuit line (Rajabzadeh et al., 2023) and SuperGrad (Wang et al., 2025) differentiate at the circuit-Hamiltonian level, SQuADDS (Shanto et al., 2024) replaces simulation by lookup in a validated design database, and Eun et al. (2023) shape-optimize transmon pads with FEM and a derivative-free global optimizer. DeviceLayout's example optimizes with PRIMA (Zhang, 2023), a reference implementation of Powell's model-based derivative-free methods, which build

an interpolation model of the objective from function values; with bounds only it runs BOBYQA (Powell, 2009).

Certified two-sided capacitance bounds. Bracketing a capacitance between complementary energy principles is classical (Specogna, 2011). Our lower bound uses the hypercircle idea of Prager and Synge (1947) with Raviart–Thomas fluxes (Raviart and Thomas, 1977) equilibrated patch by patch (Braess and Schöberl, 2007; Ern and Vohralík, 2015). The method is not new; we apply it to an island capacitance with a floating conductor on a production qubit layout and found no certified capacitance bound in the qubit-design literature.

3 Solvers, Device and Models

3.1 geode-fem

geode-fem assembles first-order Nédélec (Nédélec, 1980) and scalar P1/P2 element operators as batched $[n_{\text{elem}}, k, k]$ tensor operations on Burn. Its eigensolver is shift-invert Lanczos (Lehoucq et al., 1998) on a real symmetric pencil, factorized with faer (Kazdadi, 2026); large electrostatic solves use conjugate gradients with an algebraic-multigrid preconditioner in f64. The two codes share the formulation, the lumped junction port, the materials and the input mesh, and geode-fem follows a specification of Palace's operators (Section 8). Assembly, the eigensolver and the linear algebra are separate implementations, so agreement tests the implementation and not the shared modelling choices.

The eigenfrequency derivative used in Section 7 needs no adjoint solve, because the pencil is symmetric: for a simple eigenpair (λ, x) with $x^{\mathsf{T}} M x = 1$,

$$\frac{d\lambda}{d\theta} \;=\; x^{\mathsf{T}}\Big(\frac{\partial K}{\partial \theta} - \lambda\,\frac{\partial M}{\partial \theta}\Big)x, \tag{1}$$

the classical eigenvalue derivative of a symmetric generalized eigenproblem (Fox and Kapoor, 1968). Here K and M include the junction terms of Eq. (2); the junction does not move under the capacitor-length motion of Section 7, so those terms do not contribute. geode-fem evaluates $\partial K/\partial\theta$ and $\partial M/\partial\theta$ along a node velocity $\partial X/\partial\theta$ with a hand-written forward-mode dual-number twin of its tensor-permittivity Nédélec kernel, whose real part is tested against the assembled kernel to 10^{-10} relative. Burn's autodiff tape is not used for any derivative in this paper.

3.2 Device, mesh and materials

The device is DeviceLayout.jl v1.15.0's SingleTransmon example (AWS Center for Quantum Computing, 2025a): a transmon with a meander readout resonator and a feedline on sapphire, in a vacuum box (Figure 1). Its two design parameters are the transmon's `capacitor_length` c and the resonator's `total_length` L; the published initial design has $c = 620\,\mu$m and $L = 5000\,\mu$m. The sapphire permittivity is $\mathrm{diag}(9.3, 9.3, 11.5)$ with the optic axis normal to the chip; the configuration rotates the material axes about that normal, which leaves a uniaxial tensor unchanged, so the lab tensor is diagonal. Where a scalar model is used it is the trace average $\bar{\varepsilon} = 10.03$.

Two kinds of mesh appear. The *fixture* is an earlier generator's mesh of the initial geometry (22,684 nodes, 133,314 tetrahedra), pinned by SHA-256; Section 4 and the capacitance brackets of Section 6 use it and its refinements, which leave its polygonal surfaces unchanged. For the cross-check and the re-optimization, DeviceLayout meshes each geometry with the two parameters exposed, at first order for the ladders and at its own second order for the runs that reproduce its discretization. Its second-order mesh has the same tetrahedra with curved faces; at the same element order, curving changes the qubit by at most 0.0001% and the resonator by about 0.05%. On

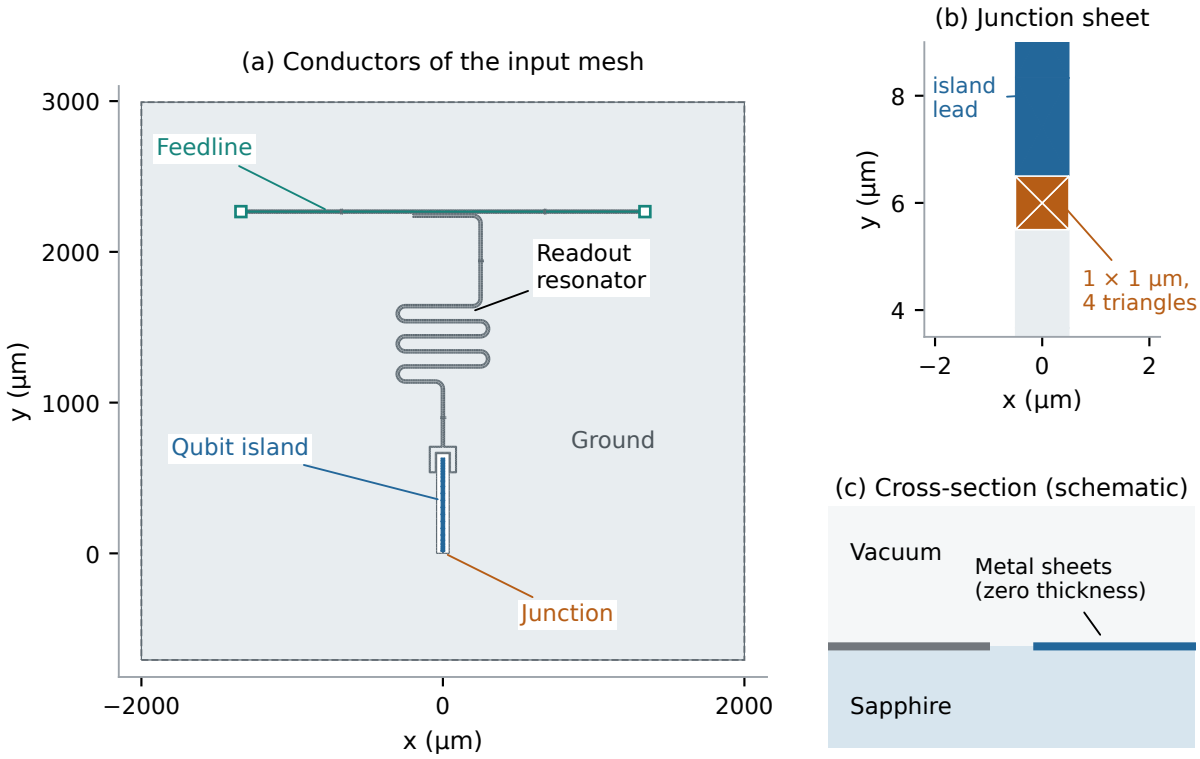


Figure 1: Benchmark geometry, drawn from the SHA-256-pinned MSH 4.1 input mesh of the DeviceLayout.jl v1.15.0 `SingleTransmon` device. (a) Top view of the three metal conductors of Section 6: the feedline (teal), the qubit island (blue), and the ground plane with the meander resonator (gray); squares mark the two readout ports. (b) The $1 \times 1\,\mu$m junction `lumped_element` sheet with its four-triangle triangulation. (c) Material cross-section, schematic and not to scale. Both solvers read this file (22,684 nodes; 133,314 tetrahedra).

the initial geometry the fixture and DeviceLayout's mesh give the same Palace qubit at h (3.4527 against 3.4520 GHz, 0.02%). PEC is applied on the metal sheets and the box; the readout ports are left open in every eigenproblem except the DeviceLayout-configuration runs of Section 5.

3.3 Junction and circuit models

The Josephson junction is a linear lumped element, $L_J = 14.860\,\text{nH}$ in parallel with $C_J = 5.5\,\text{fF}$, on the four-triangle sheet Γ of Figure 1b, as in DeviceLayout's Palace configuration. With port length ℓ and width w the uniform lumped-port construction adds

$$K_{\text{port}} = \frac{\ell}{w\,L_J}\,S_\Gamma \qquad \text{and} \qquad M_{\text{port}} = \frac{C_J\,\ell}{w}\,S_\Gamma \tag{2}$$

to the pencil $(K + K_{\text{port}})x = \omega^2(M + M_{\text{port}})x$, where S_Γ is the isotropic surface matrix on Γ. Palace's `LumpedPort` adds the same terms with the same scalings. geode-fem's rank-1 port-voltage shunt replaces S_Γ by $b\,b^{\mathsf{T}}$, with $b_i = (1/w)\int_\Gamma N_i \cdot \hat{e}\,dS$ the port-voltage functional; it keeps the physical modes to four decimals on the fixture, removes the artifact mode of Section 4, and is the junction model of every geode-fem eigensolve in Sections 5 and 7.

The electrostatic model splits the metal into three conductors: ground with the resonator, the floating feedline, and the island. Its island capacitance with the feedline floating and neutral, $C_\Sigma = C_{ii} - C_{if}^2/C_{ff}$, excludes C_J. A two-node circuit connects it to the eigenproblem: a bare qubit $1/(2\pi\sqrt{(L_J + L_s)(C_\Sigma + C_J)})$, with L_s the series inductance of the junction's current path, coupled with coefficient k to the resonator. On the fixture L_s and k were measured on three and two discretizations; extrapolated, they give bands of 0.113–0.125 nH and 0.0227–0.0310, which are estimates.

4 Same-Input-Mesh Agreement With Palace

Both codes read the fixture with the same materials and junction model at first order (Table 1). Palace's eigensolver searches above its target (read from its source at the commit of Section 9), so the six modes above 4.5 GHz come from a run with `Target` = 4.5 GHz and the qubit from one with `Target` = 2 GHz. All six agree to $\leq 0.033\%$ (worst case 0.032%), and the qubit mode to 5×10^{-5} %, with port participation 0.9942 in geode-fem and port EPR 0.99424 in Palace.

Table 1: Eigenmode agreement from the identical input mesh (22,684 nodes, 133,314 tetrahedra), first order in both solvers, junction as the lumped $L_J \parallel C_J$ shunt. geode-fem solves 156,863 Nédélec unknowns (133,108 interior); Palace refines the mesh internally around the metal sheets and solves 183,891. geode-fem frequencies are given to the archived three decimals (six for the qubit), and Δ is the archived relative difference. These are same-mesh values on a coarse mesh. Source: `benchmarks/transmon_eigen/results.toml`.

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
qubit	transmon qubit	3.452751	3.452749306805	0.00005
resonator	readout resonator	5.153	5.151335830348	0.032
mode 2	box/cavity	15.465	15.46052107794	0.029
port sheet	port-sheet artifact	17.490	17.49010903536	0.001
mode 4	box/cavity	18.693	18.69165792915	0.007
mode 5	box/cavity	20.703	20.69755679425	0.026
mode 6	box/cavity	26.088	26.08089940472	0.027

The 17.49 GHz mode is a shared artifact of the isotropic sheet term: the junction patch has one free node whose hat-function gradient carries no port voltage but still stores energy in S_Γ, with Rayleigh quotient $1/(L_J C_J)$, or $f_{LC} = 17.60$ GHz, lowered to 17.49 GHz by the adjacent volume mass; Palace gives it a port EPR of 2.49×10^{-8}. This mechanism is our inference, supported by a synthetic test; the rank-1 shunt removes the mode, and its 0.001% agreement tests only the port operator.

The fixture is coarse. Palace's own refinements of it move the qubit from 3.4527 GHz to 3.9528 GHz at $h/2$ (+14.5%) and to 4.1721 GHz at second order on h (+20.8%), and the resonator by +6.1% at $h/2$. The agreement above is code verification on one discretization; Section 5 measures the discretization error.

5 The DeviceLayout Cross-Check

The published designs. The DeviceLayout.jl blog (Peairs and Carson, 2025) reports the initial design ($c = 620\,\mu$m, $L = 5000\,\mu$m) at $[4.14, 5.591]$ GHz and the final design ($c = 1217.678\,\mu$m, $L = 7243.955\,\mu$m) at $[3.005, 3.982]$ GHz, reached by PRIMA in nine Palace runs "within 1% of our target frequencies". The example source (AWS Center for Quantum Computing, 2025b) sets the targets to $[3.0, 4.0]$ GHz, works in the variables $x_1 = \sqrt{620\,\mu\text{m}/c}$ and $x_2 = 5000\,\mu\text{m}/L$, and stops when $\text{err} = \frac{1}{2}\sum_m (f_m - t_m)^2/t_m^2 < 10^{-4}$. Its frequencies are computed with Palace on DeviceLayout's

mesh at mesh order 2 and solver order 2, with a first-order absorbing far field and two 50 Ω readout ports.

Reproduction at DeviceLayout's discretization. Palace on DeviceLayout's second-order mesh with the example's configuration file gives [4.1683, 5.6132] GHz for the initial design and [3.0164, 3.9956] GHz for the final one, 0.68% and 0.40% above the published initial pair and 0.38% and 0.34% above the final pair. The same cells with the PEC-box model used everywhere else differ from the configuration-file runs by at most 0.003%.

The ladder and its rate. For each geometry, Palace runs at order 1 on DeviceLayout's first-order mesh h and its uniform refinements $h/2$ and $h/4$ (gmsh's eight-way split, up to 12.6 million unknowns), and at order 2 on h and $h/2$; geode-fem runs at order 1 on h and $h/2$, on the same meshes. Every computed series rises under refinement (Figure 2); this is an observation, not a guarantee. At the edge of a zero-thickness metal sheet the field behaves as $E \sim r^{-1/2}$ (Meixner, 1972), so near the edge the eigenfunctions lie in H^s only for $s < 1/2$, and on uniform meshes the eigenvalue error decays as h^{2s} at any element order (Boffi, 2010): at best $O(h)$, and slower. The order-1 ladders show pre-asymptotic orders of 0.61 (qubit) and 0.70–0.73 (resonator).

Extrapolation rule. The *estimate* is the mean of two extrapolations at order 1 in the frequency, $2f_2 - f_1$: one from the order-1 pair ($h/2$, $h/4$) and one from the order-2 pair (h, $h/2$). Order 1 is the fastest rate the theory allows, so the rule makes the smallest correction consistent with it, and its estimates lean low; for the order-2 pair the order is taken from theory, because two levels give no observed order. The *band* spans six values: those two; the same pairs extrapolated at order 1 in the eigenvalue variable $\lambda \propto f^2$, $\sqrt{2f_2^2 - f_1^2}$, which set the low side; and the same pairs at the observed order, which set the high side. The band is an extrapolation range, not a bound. For the initial qubit, the circuit-model cap of Section 6 trims the band at 4.477 GHz. The rule was fixed after the static route had been computed: extrapolation at the observed order alone, the rule used before, puts the initial qubit at 4.63 GHz, above the circuit-model cap, and the observed-order values are kept only as the band's high side. The agreement of the two routes is therefore a consistency check, not a blind prediction.

Table 2: Published frequencies, Palace at DeviceLayout's discretization, and the extrapolated estimates with their bands, in GHz, from the first refinement family. Percentages are relative to the published value. Source: `benchmarks/transmon_devicelayout_crosscheck/results.toml`.

	Published	At DeviceLayout's discretization	Extrapolated estimate	Band
Initial qubit	4.14	4.1683 (+0.68%)	4.444 (+7.3%)	[4.415, 4.677]*
Initial resonator	5.591	5.6132 (+0.40%)	5.740 (+2.7%)	[5.723, 5.817]
Final qubit	3.005	3.0164 (+0.38%)	3.265 (+8.7%)	[3.236, 3.471]
Final resonator	3.982	3.9956 (+0.34%)	4.088 (+2.7%)	[4.077, 4.139]

*Under the circuit model of Section 6, the cap of 4.477 GHz trims the band to [4.415, 4.477]. The second refinement family gives 4.472 [4.458, 4.510].

Result. The initial qubit extrapolates to 4.444 GHz and the final qubit to 3.265 GHz, 7.3% and 8.7% above the published values; both resonators extrapolate to 2.7% above theirs (Table 2). Every published value lies below its band. Measured against the estimates, DeviceLayout's discretization sits 6.2% and 7.6% low on the two qubits and 2.2% and 2.3% low on the resonators. With no

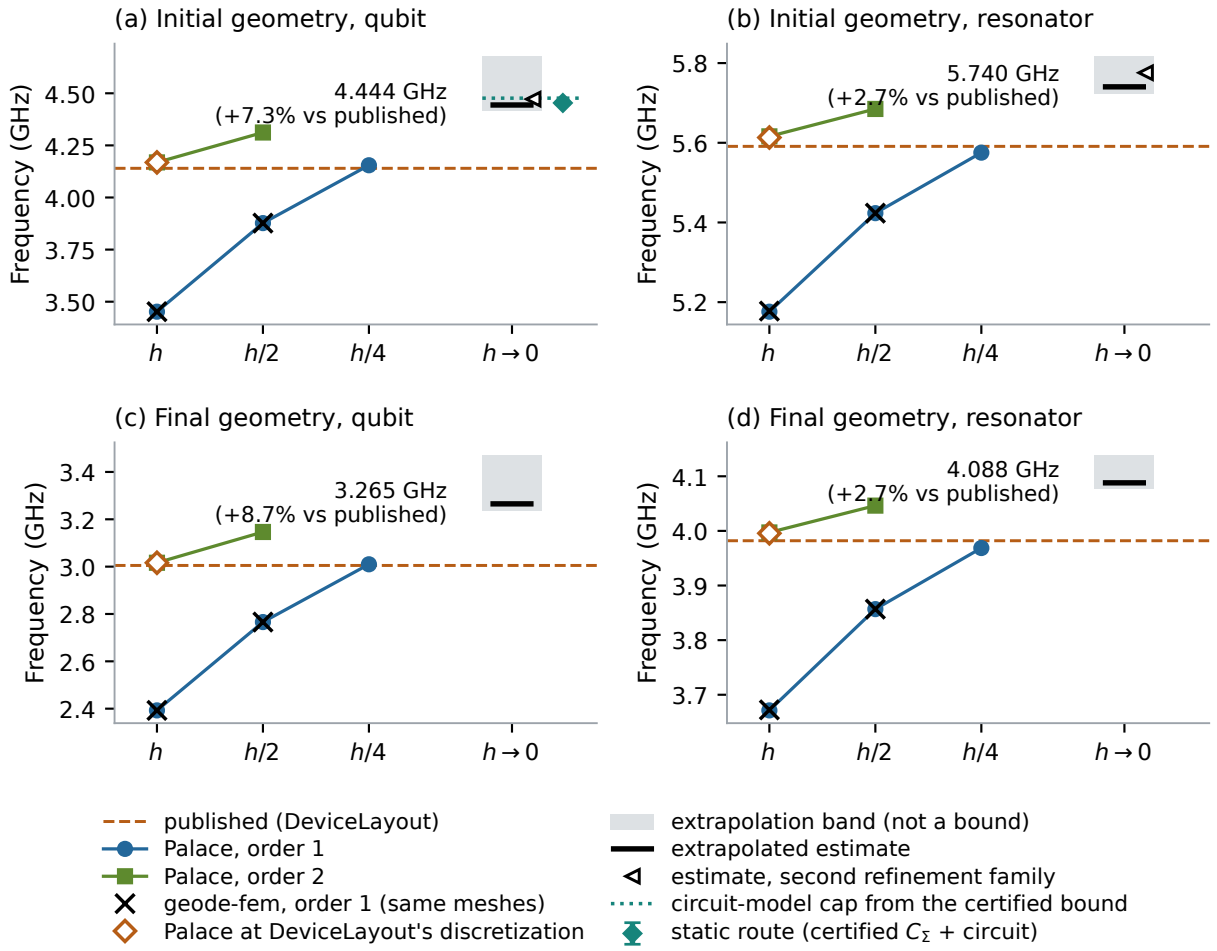


Figure 2: Resolution ladders of the two published DeviceLayout.jl SingleTransmon designs: Palace at orders 1 and 2 on DeviceLayout's mesh h and its uniform refinements, and geode-fem at order 1 on the same meshes (crosses); the open diamond is Palace at DeviceLayout's own discretization, that of the published values (dashed). The "$h \to 0$" column is categorical: the extrapolated estimate with its band and, for the initial geometry, the estimate of the second refinement family (open triangle). In (a), the static route of Section 6 and the circuit-model cap. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

extrapolation, the finest uniform levels already lie 4.1% and 4.7% above the published qubit values. The estimates rest on Palace's order-1 $h/4$ and order-2 levels, which geode-fem did not run. On the levels it did run, geode-fem and Palace agree on the qubit to 10^{-4} % on both geometries. On the resonator they differ by 0.03% at h, where Palace adds 4,946 (initial) and 5,944 (final) elements by local bisection, and by 0.0000% at $h/2$, where both codes solve the same mesh.

A second refinement family. Three checks test the initial-geometry estimate, and all three concern the qubit (Figure 3). The first is a second refinement family: Palace's own uniform refinement of the same h mesh, which first bisects the elements at the metal sheet. Its $h/2$ qubit is 3.9469 GHz against gmsh's 3.8771 GHz, and on the fixture the two refinements give 3.9528 and 3.8939 GHz, so the refinement scheme accounts for 1.5–1.8% of the $h/2$ level and the choice of h mesh for only 0.15–0.43%. The rule gives 4.472 $[4.458, 4.510]$ GHz for the qubit, at an observed

order of 0.91 against the first family's 0.61, and 5.776 [5.768, 5.800] GHz for the resonator. The two estimates lie 0.64% apart and the second band lies inside the first, so we quote the initial qubit as 4.444–4.472 GHz, 7.3–8.0% above the published value.

Adaptive refinement. A further check comes from Palace's adaptive mesh refinement, run from the same h mesh at order 2, with the error indicator summed over the qubit and resonator modes, Dörfler marking at 0.7 and nonconformal refinement. In five solves from 0.95 to 11.0 million unknowns the qubit rises from 4.1683 to 4.4297 GHz, 0.32% below the estimate and inside the band. The third solve, at 2.2 million unknowns, already exceeds the finest uniform level (4.3118 GHz at 7.0 million). The run has not reached a plateau (the last step moved the qubit by +0.55%; 3.1 h, 93 GiB peak on 32 ranks). A three-point fit $f_\infty - CN^{-\alpha}$ through the last three solves gives 4.4552 GHz ($\alpha = 0.80$), but it is exactly determined and ill-conditioned: +0.1% on the finest level moves it by 24 MHz. The adaptive resonator rises more slowly under this marking, and its finest level, 5.6986 GHz, is still below its band, so the adaptive run checks the qubit estimate only.

A common-limit fit. The last check fits the order-1 ($h/2$, $h/4$) and order-2 (h, $h/2$) series of each ladder with one common limit and order. This fit is exactly determined; it lies 0.5–1.3% above the rule's estimate on every first-family series, as the rule's leaning low predicts, and +0.1% on the order-2 $h/2$ level alone moves the initial-qubit fit by 20 MHz. Section 7 repeats the second family on the re-optimized design.

6 A Static Route: Certified Island Capacitance

The certificate. For the reduced electrostatic problem (island at potential 1, ground and resonator at 0, feedline floating with zero net charge), the potential ϕ satisfies $0 \le \phi \le 1$ by the maximum principle, and $C_\Sigma = \int \varepsilon \nabla\phi \cdot \nabla\phi$ with $\varepsilon = \varepsilon_0\varepsilon_r$. Every conforming finite-element energy is an upper bound on C_Σ, because C_Σ is the minimum of the energy over admissible potentials. For the lower bound, take any piecewise polynomial field D on the mesh cut along the zero-thickness sheets. Integrating by parts and using $0 \le \phi \le 1$ on every term gives

$$\int D \cdot \nabla\phi \;\ge\; N(D) \;=\; Q_{\text{island}}(D) - |Q_{\text{feed}}(D)| - \|\operatorname{div} D\|_{L^1} - \|[D \cdot n]\|_{L^1(\mathcal{F})}, \tag{3}$$

where Q are the fluxes of D into the two conductors and $[D \cdot n]$ is the normal jump on the interior faces $\mathcal{F}$ off the sheets. Cauchy–Schwarz, $\left(\int D \cdot \nabla\phi\right)^2 \le \int D \cdot \varepsilon^{-1} D \int \varepsilon \nabla\phi \cdot \nabla\phi$, then gives, whenever $N(D) > 0$,

$$C_\Sigma \;\ge\; \frac{N(D)^2}{\int D \cdot \varepsilon^{-1} D}. \tag{4}$$

The bound holds for any D, so solver tolerance can only weaken it. We take D from the P2 potential, equilibrated patch by patch into Raviart–Thomas fluxes of order one to three, with an independent Crouzeix–Raviart flux as a check. The construction is validated against an exact parallel plate and an analytic coaxial bracket, and mutation tests reject a grounded-feedline or jump-free certificate. *Certified* in this paper means rigorous for the discrete polygonal geometry of the mesh, with lower bounds floored and upper bounds ceiled. The certificate is evaluated in plain f64, not interval arithmetic; the worst-case summation error is about 10^{-10} relative, far below the rounding. Refinement does not move the polygonal surfaces, so all meshes of one geometry bound the same exact capacitance, and a family bracket is the largest lower bound and the smallest upper bound.

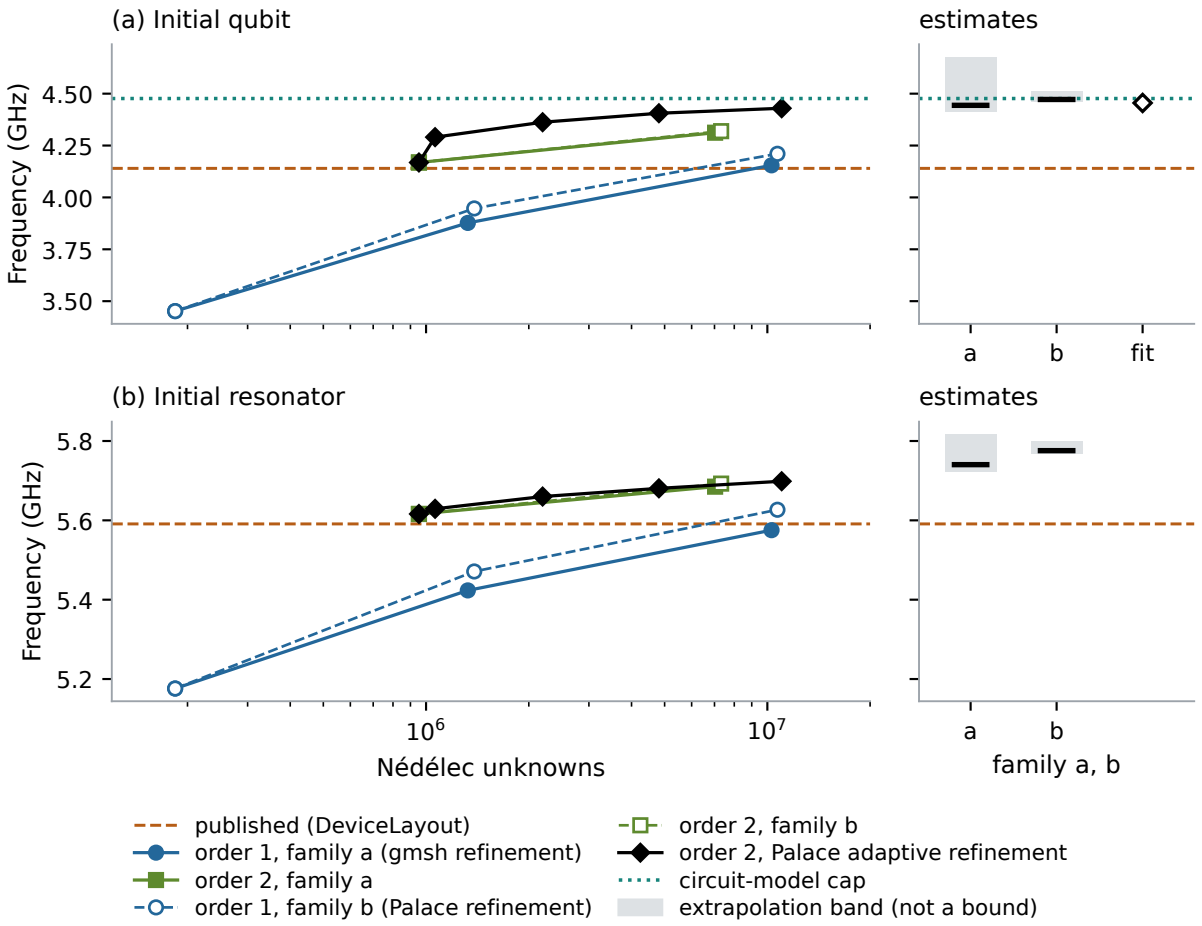


Figure 3: Checks of the extrapolated estimate on the initial geometry. Left: every Palace level against the number of Nédélec unknowns, for the two refinement families of DeviceLayout's mesh h (filled: gmsh's eight-way split, the ladder of Figure 2; open: Palace's refinement) and Palace adaptive refinement from h at order 2 (diamonds). Right, on the same frequency axis: the estimates of the two families with their bands and, for the qubit only, the adaptive run's ill-conditioned three-point fit; the resonator fit ($\alpha = 0.25$) is not reliable and not shown. In (a), the circuit-model cap. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

The unmodified device. Uniform refinement of the fixture with P1 and P2 elements gives the upper bounds of Figure 4: with the scalar model, from 137.706752 fF at P1 on h to 83.015285 fF at P2 on $h/2$, and with the tensor from 136.684731 to 85.008425 fF. On adaptively refined meshes the certificate closes the gap: the island capacitance of the unmodified initial design lies in $[77.40, 77.61]$ fF with the scalar $\bar{\varepsilon}$ and in $[79.56, 79.76]$ fF with the sapphire tensor (Figure 4). The certified tensor-minus-scalar difference, computed from the unrounded bounds, is 1.95–2.35 fF, or 2.5–3.1% of the scalar value. The fixture itself overstates the tensor C_Σ by at least 56.92 fF, which is at least 71.3% of the exact value, and the scalar one by at least 77.4%.

From capacitance to the qubit frequency. Reading the certified tensor bracket through the two-node circuit of Section 3.3, over every corner of the C_Σ, L_s and k bands, with the resonator at its extrapolated estimate, gives the initial dressed qubit in $[4.449, 4.458]$ GHz. That range lies inside the eigenmode band of Section 5 and inside the range of the two family estimates, and the first

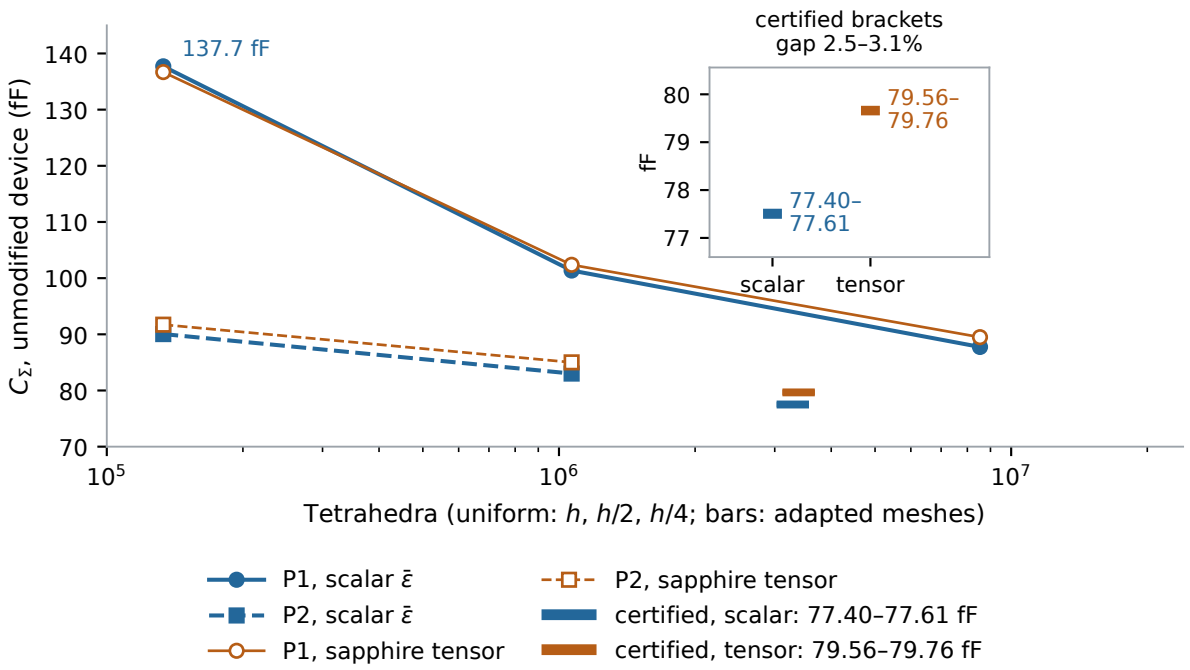


Figure 4: Mesh convergence of the island capacitance C_Σ of the unmodified device under uniform refinement of the fixture, with the scalar trace-averaged permittivity (blue) and the sapphire tensor (orange). Every marker is a conforming upper bound on the exact capacitance of the meshed polygonal geometry. The short bars are the certified two-sided brackets from adapted meshes, each drawn at the tetrahedron count of the mesh that sets its lower bound (3,289,543 and 3,388,197) and magnified in the inset on a fF scale; the inset title gives the certified tensor-minus-scalar gap. Sources: `c_sigma_convergence.toml`, `c_sigma_bracket.toml` and `c_sigma_bracket_tensor.toml`.

family's estimate is 0.22% below its midpoint. Because the extrapolation rule was chosen after this range was known, the 0.22% is a consistency check, not an independent prediction. The range is an estimate, because L_s and k are.

A model-dependent cap. Under the circuit model, a series inductance and a coupling to a higher mode can only lower the qubit, so the certified lower bound $C_\Sigma \geq 79.56\,\mathrm{fF}$ caps it at $1/(2\pi\sqrt{L_J(C_\Sigma + C_J)}) = 4.477\,\mathrm{GHz}$. The cap is rigorous only given that model (one island node with $L_J \parallel C_J$, $L_s \geq 0$, coupling only to modes above the qubit, a quasi-static C_Σ) and rests on the fixture, not on the ladder meshes, which agree with it to 0.02% at h, so we list it as model-dependent, not certified. It trims the band's observed-order high side (4.677 GHz) and does not depend on the extrapolation rule. Every computed level and both family estimates lie below it. The first family's common-limit fit, 4.4802 GHz, lies 3.6 MHz (0.08%) above it, an excess equal to a 0.019% change of one order-2 level and so within that fit's conditioning.

7 Re-Optimization and Its Controls

The published final design misses its targets at the extrapolated frequencies. We repeat DeviceLayout's optimization against its own discretization, to compare optimizers in one Palace environment, and against the extrapolated frequencies, to find the design that meets the targets there.

The capacitor_length derivative. DeviceLayout keeps the junction fixed and builds the island upward from it, so a change of c translates every node above the island rigidly and stretches the island band. geode-fem turns this motion into a node velocity $\partial X/\partial c$, checks that every metal outline node in a stretched band moves along its own edge (3,146 nodes on the initial mesh), and evaluates Eq. (1) with the sapphire tensor for both modes. The contraction takes 0.44 s. On the initial mesh it matches a central difference of two re-solves of the moved mesh ($\pm 0.5\,\mu$m) to every printed digit: -2.443227×10^{-3} GHz/μm for the qubit and -3.964×10^{-6} GHz/μm for the resonator.

The total_length column. DeviceLayout re-lays the meander when L changes, so no fixed-topology node motion represents it. A transmission-line model holds $f_r(L + \delta)$ constant and takes $\partial f_q/\partial L = 0$, with $\delta = 216\,\mu$m from one finite difference of two geode-fem order-1 solves ($L = 5000$ and $5250\,\mu$m). δ is weakly determined: remesh noise of about 0.15% in either f_r moves it by about $\pm 175\,\mu$m. The model enters only the Jacobian, and every iterate is evaluated on the true objective, so a poorer Jacobian costs steps, not accuracy.

Newton on the high-fidelity objective. DeviceLayout re-meshes every design, so the first-order objective is not smooth in the parameters: from $c = 620$ to $650\,\mu$m the remeshed finite difference of the qubit frequency is -5.23×10^{-3} GHz/μm, against a fixed-topology derivative of -2.28×10^{-3}. The iteration therefore runs Newton directly on the high-fidelity objective F, in DeviceLayout's variables (x_1, x_2), in which the frequencies are close to linear and nearly decoupled: a probe in either variable moves its own mode by 4.5% and the other by at most 0.05%. The c column of the Jacobian is geode-fem's first-order derivative scaled by the fidelity ratio $S = F/f_h$ at the same design. Each step costs one geode-fem first-order solve with its gradient (19–25 s) and one high-fidelity evaluation; the one-time δ calibration is not counted.

Table 3: Palace runs to DeviceLayout's criterion (err $< 10^{-4}$) at its discretization, counting the initial design as run 1, all in our Palace environment (build, mesher and platform) except the published PRIMA run. The Newton variants use the transmission-line L column with the stated δ; Broyden's count includes its two probes. Sources: `benchmarks/transmon_devicelayout_optimize/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

Method	c column	δ from	Runs	First-step err	Design at criterion (μm)
PRIMA, published	none	none	9	unpublished	(1217.678, 7243.955)
PRIMA, our environment	none	none	10	n/a	(1209.125, 7207.551)
Newton	shape derivative	geode-fem	3	7.9×10^{-4}	(1225.346, 7223.250)
Newton	scaling law	geode-fem	3	1.7×10^{-4}	(1232.664, 7221.270)
Newton	scaling law	Palace	3	2.1×10^{-4}	(1229.026, 7225.712)
Broyden	two probes	none	5	6.2×10^{-4}	(1224.847, 7232.447)

Against DeviceLayout's discretization. Table 3 compares the optimizers on one objective. Newton with the shape-derivative column meets the criterion at its second step, after 3 Palace runs, and its three geode-fem solves add about 70 s, against about 4 min per Palace run on 32 ranks; a third step reaches err $= 1.4 \times 10^{-7}$ at $(1236.360, 7224.900)\,\mu$m. Two gradient-free controls test where the 3 runs come from. The first replaces the c column by the scaling law $\partial F_q/\partial x_1 = F_q/x_1$, which follows from $f_q \propto 1/\sqrt{C_\Sigma} \propto x_1$. With the geode-fem δ it also takes 3 runs, and its first step has an err 4.7 times lower (1.7×10^{-4} against 7.9×10^{-4}), about 2.2 times closer in frequency. With $\delta = 563.879\,\mu$m taken from a Palace probe instead, so that no geode-fem input remains, it again takes 3. The second control, Broyden's method (Broyden, 1965) from two forward finite-difference

probes, takes 5. DeviceLayout's own PRIMA call, re-run verbatim in our environment, takes 10: runs 1–5 are its initial interpolation set of $2n + 1 = 5$ points (Powell, 2009), and run 9 misses the criterion by 15% (err $= 1.15 \times 10^{-4}$). In this environment the published final design already meets the criterion (err $= 1.5 \times 10^{-5}$), so the environment moves PRIMA's trajectory, not the attainability of the target; the published count of 9 is from DeviceLayout's environment. In one environment, then, the comparison is 3 Newton runs against 10.

Why the shape-derivative step lands farther. At the initial design the scaled shape derivative gives $\partial F_q/\partial x_1 = 3.66\,\mathrm{GHz}$, within 1.5% of the 3.71 GHz of Broyden's Palace probe, so the derivative measures the local slope well. Both local slopes are 11–12% shallower than the scaling law's $F_q/x_1 = 4.168\,\mathrm{GHz}$, because the frequency is not exactly proportional to x_1 near the initial design, and the first step is long (x_1 from 1 to about 0.7). A local tangent therefore overshoots: the shape-derivative step lands at $c = 1338.338\,\mu\mathrm{m}$ and Broyden's at $1320.522\,\mu\mathrm{m}$, against about $1230\,\mu\mathrm{m}$ at the criterion, while the scaling law, which follows $f_q \propto x_1$ over the whole step, lands at $1196.933\,\mu\mathrm{m}$.

Against the extrapolated frequencies. Starting from the published final design, each Newton step uses the extrapolated estimate and the geode-fem derivative at the current design, and each new design is evaluated by a full ladder with the rule of Section 5 (Table 4). The first overshoots the qubit by −1.09%; the second lands at +0.16% and −0.04% of the targets, with both targets inside the bands. On both ladders geode-fem and Palace agree on the qubit to 2×10^{-4} %. One scaling-law step from the same start, evaluated by the Palace half of the ladder, lands 3.6 and $8.2\,\mu\mathrm{m}$ short of step 2 and at +0.13% and +0.12% of the targets: one gradient-free step does what the two shape-derivative steps did. Each ladder took 1.4 and 1.6 h of Palace wall time, dominated by the order-2 $h/2$ cell.

Table 4: Steps against the extrapolated frequencies, in GHz. Each design is evaluated by a full resolution ladder (the scaling-law step and the second family by Palace alone) and the rule of Section 5; values are estimates with their bands. Family b refines step 2's h mesh with Palace's uniform refinement instead of gmsh's. Sources: `benchmarks/transmon_devicelayout_optimize/results.toml`, `benchmarks/transmon_devicelayout_controls/results.toml` and, for the published final design, `benchmarks/transmon_devicelayout_crosscheck/results.toml`.

Design	c (μm)	L (μm)	f_q [band]	f_r [band]
Published final	1217.678	7243.955	3.265 [3.236, 3.471]	4.088 [4.077, 4.139]
Step 1	1481.657	7407.945	2.967 [2.938, 3.188]	4.004 [3.993, 4.055]
Step 2	1446.078	7416.352	3.005 [2.979, 3.221]	3.998 [3.987, 4.067]
Step 2, family b	1446.078	7416.352	3.020 [3.003, 3.101]	4.022 [4.016, 4.043]
Scaling-law step	1442.479	7408.181	3.004 [2.977, 3.235]	4.005 [3.994, 4.055]

The re-optimized design. Step 2 is the design whose extrapolated frequencies sit on the targets: $c = 1446.078\,\mu\mathrm{m}$, 18.8% longer than published, with a resonator 2.4% longer ($+172\,\mu\mathrm{m}$). The second refinement family at step 2 gives 3.0203 [3.0033, 3.1014] GHz for the qubit and 4.0225 GHz for the resonator, 0.52% and 0.60% above the first family, the same spread as on the initial geometry; both targets lie just below its band, so on family b step 2 is slightly short. Mapped to 3.0 GHz by the scaling law $f_q \propto c^{-1/2}$, the two families' qubit estimates call for $c = 1451$ and $1466\,\mu\mathrm{m}$, +19.1% and +20.4% over the published length. Mapped linearly through $dF/dc \approx S\,\partial f_h/\partial c = -9.05 \times 10^{-4}\,\mathrm{GHz}/\mu\mathrm{m}$, the slope at the step-1 design where step 2 started, and centred on the 3.0 GHz target, the first family's qubit band corresponds to c between about 1423 and $1690\,\mu\mathrm{m}$, about +17% to +39%; under

the scaling law the same band gives +17.1% to +36.9%, and it contains family b's band (+19.0% to +26.9%). Since the estimates lean low, the common-limit fit, which puts step 2 at 3.039 and 4.019 GHz, would call for a longer capacitor still.

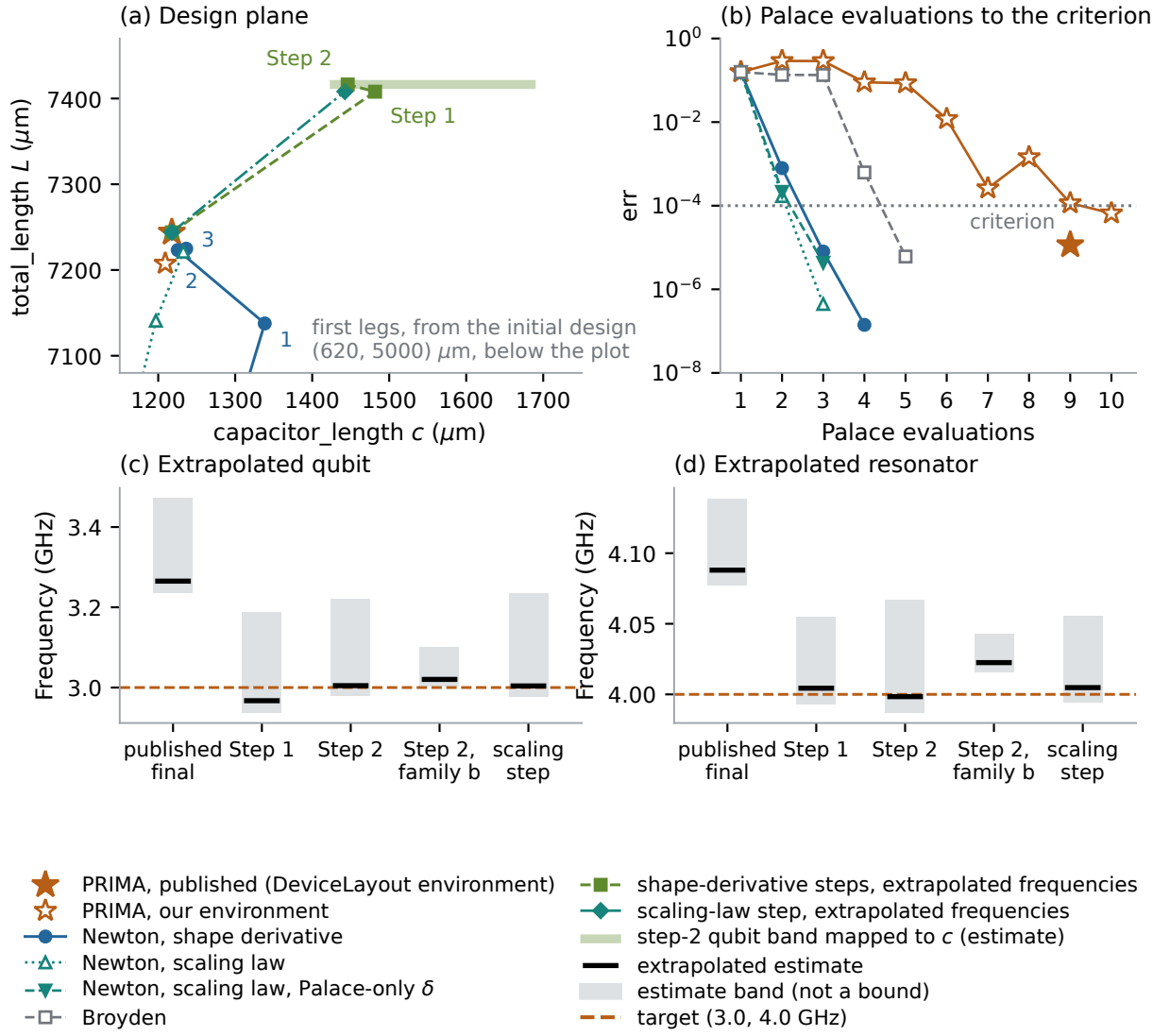


Figure 5: The re-optimization and its controls. (a) The design plane near the optimum: the published final design, PRIMA's end point in our environment, the Newton iterates against DeviceLayout's discretization (numbered: shape derivative; the first legs start off the plot), and the steps against the extrapolated frequencies, with the step-2 qubit band mapped linearly to capacitor_length around the 3.0 GHz target (an estimate). (b) DeviceLayout's err against the number of Palace evaluations at its discretization, in our environment; PRIMA's published end point is plotted at its 9 runs. This counts Palace evaluations, not total cost: each shape-derivative Newton point also costs one geode-fem solve. (c, d) Extrapolated estimates and bands against the targets, with step 2 in both refinement families. Sources: the `optimize`, `controls` and `crosscheck` artifacts under `benchmarks/transmon_devicelayout_*/`.

8 How geode-fem Was Designed and Built

geode-fem implements the whiteroom L1–L4 operator specification of Crutcher Dunnavant (Dunnavant, 2026). He also gave design input on the solver's operator layer and contributed code: the migration to Rust 2024, the Metal (Apple GPU) backend, the present layout of the core crate and the examples, and the geode-util crate. The rest of the code, the experiments and the text of this paper were produced by frontier AI models working as role-separated agents: Anthropic's Claude (Opus, Fable and Sonnet models, through Claude Code) and OpenAI's GPT Sol and GPT Astra models (via Codex). The agents were coordinated by Loom (Walters, 2025), an issue, worktree and pull-request workflow, and by a shared Squad room. Nearly every change was reviewed by a separate agent session. The reviews of this paper recomputed its numbers from the archived artifacts, reproduced runs, and mutation-tested the scripts that check the text and the figures against those artifacts. The human author directed the research, made every scoping and claim decision, and takes responsibility for the content.

9 Reproducibility

Every result in this paper is read from an archived, scripted artifact.[1] Each table and figure caption names its artifact; the circuit terms L_s and k come from `benchmarks/transmon_loop_inductance/` and `benchmarks/transmon_coupling/`, the Palace cells are under `reference/fixtures/transmon_palace/`, and the DeviceLayout meshes are generated by `reference/julia/generate_transmon_param.jl`. The generated artifacts are pinned by regression tests. The versions are geode-fem commit `3174015` for the same-mesh agreement, with the commit of each later result recorded in its artifact, Palace `fba6a5b`, DeviceLayout.jl v1.15.0, PRIMA.jl 0.2.4 and Burn 0.21. The DeviceLayout ladders, the adaptive run and the controls ran on AWS Graviton4 hosts (`r8g.8xlarge`, 32 vCPU, and `r8g.4xlarge`, 16 vCPU); the capacitance brackets ran on a 28-core, 96 GiB workstation. geode-fem's direct solves at the ladders' $h/2$ peaked at 94 and 136 GB, which is why only Palace ran the larger cells. The Palace runs consume an MSH 2.2 conversion of each mesh, because MFEM rejects DeviceLayout's multi-block MSH 4.1 file; the conversion keeps every node and physical group and reproduces the eigenvalues bit for bit. The figure scripts in `figures/src/` read the artifacts directly.

10 Discussion

What is established, and how. Table 5 sorts the results by the kind of evidence behind them. The brackets are the only bounds, and the cap follows from a bracket only under a circuit model. What stands without any extrapolation is the adaptive run's 4.4297 GHz, 7.0% above the published initial qubit, and the finest uniform levels, 4.1% and 4.7% above the two published qubits; under the monotone rise every series shows, these are lower estimates.

Why the shape derivative gives no advantage here. The derivative is a central tool of the re-optimization, but three properties of this benchmark remove its advantage. First, the problem has two nearly decoupled parameters, close to linear in DeviceLayout's variables, so Newton's method converges with any reasonable Jacobian: the shape derivative, a two-probe secant, or the scaling law $f_q \propto x_1$, which costs nothing. Second, a finite-difference gradient costs only $N + 1 = 3$ solves. Third, the coarse-mesh, fixed-topology derivative is the slope of a different discrete problem from the one the iteration evaluates: DeviceLayout remeshes every design and Palace evaluates it at second order, so a remeshed finite difference of that objective is the more faithful slope; here the

[1] The geode-fem source is public at https://github.com/rjwalters/geode-fem. Archival DOI to be added at deposit.

Table 5: Evidence tiers. Certified: a guaranteed bound for the meshed polygonal geometry. Model-dependent: a rigorous consequence of a certified bound under a stated circuit model. Measured: a solve on a stated discretization. Estimated: an extrapolation with a stated rule. Consistency: an agreement that is not independent evidence.

Tier	Results
Certified	C_Σ of the unmodified device: $[77.40, 77.61]$ fF (scalar), $[79.56, 79.76]$ fF (tensor); tensor-minus-scalar gap 2.5–3.1%
Model-dependent	The initial-qubit cap of 4.477 GHz: two-node circuit, quasi-static C_Σ, bracket certified on the fixture's polygonal geometry, not on the ladder meshes
Measured	Same-mesh agreement (Table 1); Palace at DeviceLayout's discretization within 0.7% of the published pairs; every ladder and adaptive level (adaptive initial qubit +7.0%, finest uniform levels +4.1% and +4.7%); the gradient check; Palace runs to DeviceLayout's criterion in one environment: 3 for each Newton variant, 5 for Broyden's method, 10 for PRIMA
Estimated	Extrapolated frequencies and bands (Table 2), the initial qubit 4.444–4.472 GHz over two refinement families; the adaptive fit and the common-limit fits; the static-route qubit range; L_s and k; the re-optimized capacitor, +19 to +20% over two families and about +17 to +39% within the band
Consistency	The two routes agree to 0.22% on the initial qubit; the extrapolation rule was fixed after the static route was known, so this is not a blind prediction
Not claimed	Agreement with measurement; a verified hit of 3.0/4.0 GHz in the $h \to 0$ limit; a global optimum; an advantage of the shape derivative on this problem; eigenmode-participation (EPR) derivatives; a speed advantage over Palace

scaled derivative lies 1.5% below one. That the derivative is exact for its own discrete problem and cheap (0.44 s) does not change the number of Palace runs.

Derivatives pay off in other regimes. With N parameters, a discrete adjoint gives all N derivatives of one objective for about one extra solve, where finite differences need N more; coupled targets with no per-parameter scaling law need a computed Jacobian; and freeform shapes have many parameters by construction. A companion study (Walters, 2026) applies geode-fem's discrete shape adjoint to a conformal patch antenna with 38 shape parameters; its comparison with gradient-free search is still being run, so it does not yet show whether the derivative helps there.

Threats and limitations. Agreement between two codes is evidence against implementation error, not against shared modelling choices such as the port sheet behind the 17.49 GHz mode. The extrapolations assume the rate the sheet-edge singularity allows, their high sides are wide, and the adaptive run had not reached a plateau. The circuit route uses a bracket certified on a different mesh of the same initial geometry, and L_s and k were extrapolated from two or three discretizations. The certificate is evaluated in f64 and validated by tests, not formally verified. The estimates are for DeviceLayout's geometry under a lossless, linear model ($L_J \parallel C_J$, open readout ports in the ladders, zero-thickness metal), not for a fabricated device, and only one device is studied. The eigenmode branch is not yet differentiated with respect to participation ratios, which needs eigenvector derivatives (Nelson, 1976).

What a practitioner can do. The slow rate comes from the sheet edges, and the standard remedy for an edge singularity is to refine toward it, with a mesh graded toward the edges or with hp refinement (Demkowicz, 2006), which recover faster convergence than uniform refinement. Palace's adaptive run is a measured instance: at 2.2 million unknowns it already exceeds the finest uniform level at 7.0 million.

11 Conclusion

Palace reproduces the published frequencies of the DeviceLayout.jl SingleTransmon example at the example's own discretization, and that discretization sits 6–8% low on the qubits. An adaptive Palace run puts the initial qubit 7.0% above its published value with no extrapolation, and the extrapolated qubit-frequency estimates are 7–9% above both published values; the zero-thickness sheet edge limits uniform refinement to at best $O(h)$, so these estimates lean low. geode-fem reproduces Palace's qubit on every level both codes ran, and a certified bracket on the island capacitance, read through a circuit model, is consistent with that shift and caps the initial qubit under that model. At the extrapolated frequencies the 3.0/4.0 GHz design needs a capacitor about 19–20% longer than published, and Newton's method in DeviceLayout's variables reaches the example's criterion in 3 Palace runs against PRIMA's 10, with or without the shape derivative. For design examples that are meant to be copied, the discretization is part of the result and belongs in the report beside it.

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References

Volkan Akçelik, George Biros, Omar Ghattas, David Keyes, Kwok Ko, Lie-Quan Lee, and Esmond G Ng. Adjoint methods for electromagnetic shape optimization of the low-loss cavity for the International Linear Collider. *Journal of Physics: Conference Series*, 16:435–445, 2005. doi: 10.1088/1742-6596/16/1/059.

Robert Anderson, Julian Andrej, Andrew Barker, Jamie Bramwell, Jean-Sylvain Camier, Jakub Cerveny, Veselin Dobrev, Yohann Dudouit, Aaron Fisher, Tzanio Kolev, Will Pazner, Mark Stowell, Vladimir Tomov, Ido Akkerman, Johann Dahm, David Medina, and Stefano Zampini. MFEM: a modular finite element methods library. *Computers & Mathematics with Applications*, 81:42–74, 2021. doi: 10.1016/j.camwa.2020.06.009.

AWS Center for Quantum Computing. DeviceLayout.jl: computer-aided design of quantum integrated circuits. https://github.com/aws-cqc/DeviceLayout.jl, 2025a. v1.15.0 used for geometry and mesh generation in this work.

AWS Center for Quantum Computing. Single-Transmon Simulation · DeviceLayout.jl. DeviceLayout.jl v1.15.0 documentation and `examples/SingleTransmon/SingleTransmon.jl`, https://aws-cqc.github.io/DeviceLayout.jl/stable/examples/singletransmon/, 2025b. Source of the optimization targets, variables and stopping criterion, and of the discretization settings.

Daniele Boffi. Finite element approximation of eigenvalue problems. *Acta Numerica*, 19:1–120, 2010. doi: 10.1017/S0962492910000012.

Dietrich Braess and Joachim Schöberl. Equilibrated residual error estimator for edge elements. *Mathematics of Computation*, 77(262):651–672, 2007. doi: 10.1090/s0025-5718-07-02080-7.

C. G. Broyden. A class of methods for solving nonlinear simultaneous equations. *Mathematics of Computation*, 19 (92):577–593, 1965. doi: 10.1090/S0025-5718-1965-0198670-6.

Leszek Demkowicz. *Computing with hp-Adaptive Finite Elements, Volume 1: One and Two Dimensional Elliptic and Maxwell Problems*. Chapman and Hall/CRC, 2006. doi: 10.1201/9781420011685.

Crutcher Dunnavant. palace_whiteroom. https://github.com/crutcher/palace_whiteroom, 2026. Software repository. Accessed 2026-10-13.

Axel M. Eriksson, Lukas J. Splitthoff, Harsh Vardhan Upadhyay, Pietro Campana, Niranjan Pittan Narendiran, Kunal Helambe, Linus Andersson, and Simone Gasparinetti. Automated, physics-guided, multi-parameter design optimization for superconducting quantum devices, 2025. URL https://arxiv.org/abs/2508.18027.

Alexandre Ern and Martin Vohralík. Polynomial-degree-robust a posteriori estimates in a unified setting for conforming, nonconforming, discontinuous galerkin, and mixed discretizations. *SIAM Journal on Numerical Analysis*, 53(2): 1058–1081, 2015. doi: 10.1137/130950100.

Sungjun Eun, Seong Hyeon Park, Kyungsik Seo, Kibum Choi, and Seungyong Hahn. Shape optimization of superconducting transmon qubits for low surface dielectric loss. *Journal of Physics D: Applied Physics*, 56(50): 505306, 2023. doi: 10.1088/1361-6463/acf7cf.

R. L. Fox and M. P. Kapoor. Rates of change of eigenvalues and eigenvectors. *AIAA Journal*, 6(12):2426–2429, 1968. doi: 10.2514/3.5008.

Sarah Quiñones El Kazdadi. faer: A linear algebra library for the Rust programming language. *Journal of Open Source Software*, 11(123):6099, 2026. doi: 10.21105/joss.06099.

Richard B. Lehoucq, Danny C. Sorensen, and Chao Yang. *ARPACK Users' Guide: Solution of Large-Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods*. SIAM, 1998. doi: 10.1137/1.9780898719628.

Josef Meixner. The behavior of electromagnetic fields at edges. *IEEE Transactions on Antennas and Propagation*, 20 (4):442–446, 1972. doi: 10.1109/TAP.1972.1140243.

Jean-Claude Nédélec. Mixed finite elements in $\mathbb{R}^3$. *Numerische Mathematik*, 35(3):315–341, 1980. doi: 10.1007/BF01 396415.

Richard B. Nelson. Simplified calculation of eigenvector derivatives. *AIAA Journal*, 14(9):1201–1205, 1976. doi: 10.2514/3.7211.

N.K. Nikolova, J.W. Bandler, and M.H. Bakr. Adjoint techniques for sensitivity analysis in high-frequency structure CAD. *IEEE Transactions on Microwave Theory and Techniques*, 52(1):403–419, 2004. doi: 10.1109/tmtt.2003.8209 05.

William L. Oberkampf and Christopher J. Roy. *Verification and Validation in Scientific Computing*. Cambridge University Press, 2010. doi: 10.1017/cbo9780511760396.

Palace developers, AWS Center for Quantum Computing. Palace: PArallel LArge-scale Computational Electromagnetics. https://github.com/awslabs/palace, 2023. Open-source 3D finite-element solver; git fba6a5b used in this work.

Greg Peairs and Hugh Carson. Design quantum integrated circuits with open-source software DeviceLayout.jl from AWS. https://aws.amazon.com/blogs/quantum-computing/design-quantum-integrated-circuits-with-open-source-software-devicelayout-jl-from-aws/, 2025. AWS Quantum Technologies Blog; source of the SingleTransmon designs and their published frequencies.

M. J. D. Powell. The BOBYQA algorithm for bound constrained optimization without derivatives. Technical Report DAMTP 2009/NA06, Department of Applied Mathematics and Theoretical Physics, University of Cambridge, 2009.

W. Prager and J. L. Synge. Approximations in elasticity based on the concept of function space. *Quarterly of Applied Mathematics*, 5(3):241–269, 1947. doi: 10.1090/qam/25902.

Taha Rajabzadeh, Zhaoyou Wang, Nathan Lee, Takuma Makihara, Yudan Guo, and Amir H. Safavi-Naeini. Analysis of arbitrary superconducting quantum circuits accompanied by a Python package: SQcircuit. *Quantum*, 7:1118, 2023. doi: 10.22331/q-2023-09-25-1118.

P. A. Raviart and J. M. Thomas. A mixed finite element method for 2-nd order elliptic problems. In *Mathematical Aspects of Finite Element Methods*, volume 606 of *Lecture Notes in Mathematics*, pages 292–315. Springer, 1977. doi: 10.1007/bfb0064470.

P. J. Roache. Quantification of uncertainty in computational fluid dynamics. *Annual Review of Fluid Mechanics*, 29 (1):123–160, 1997. doi: 10.1146/annurev.fluid.29.1.123.

Sadman Shanto, Andre Kuo, Clark Miyamoto, Haimeng Zhang, Vivek Maurya, Evangelos Vlachos, Malida Hecht, Chung Wa Shum, and Eli Levenson-Falk. SQuADDS: A validated design database and simulation workflow for superconducting qubit design. *Quantum*, 8:1465, 2024. doi: 10.22331/q-2024-09-09-1465.

David Sommers, Zach Degnan, Divita Gautam, Yi-Hsun Chen, Chun-Ching Chiu, Arkady Fedorov, and Prasanna Pakkiam. Open-source highly parallel electromagnetic simulations for superconducting circuits, 2025. URL https://arxiv.org/abs/2511.01220.

Ruben Specogna. Complementary geometric formulations for electrostatics. *International Journal for Numerical Methods in Engineering*, 86(8):1041–1068, 2011. doi: 10.1002/nme.3089.

Tracel AI. Burn: a comprehensive deep-learning and tensor-computation framework in Rust. https://github.com/tracel-ai/burn, 2024. v0.21.0 used. No journal or arXiv paper; cited as software.

Robb Walters. Loom: AI-powered development orchestration. https://github.com/rjwalters/loom, 2025. Software repository. Accessed 2026-10-13.

Robb Walters. Shape-adjoint design of a curved conformal 2.4 GHz patch antenna on a fixed 1.6 mm board with a differentiable open-radiator FEM, 2026. Manuscript in preparation, `papers/conformal-antenna-diffopt/` in https://github.com/rjwalters/geode-fem.

Ziang Wang, Feng Wu, Hui-Hai Zhao, Xin Wan, and Xiaotong Ni. SuperGrad: a differentiable simulator for superconducting processors. *Quantum*, 9:1722, 2025. doi: 10.22331/q-2025-04-24-1722.

Jiale Ye, Jiaheng Wang, and Yu-xi Liu. Electromagnetic feature extraction in superconducting quantum circuits: An open-source finite-element workflow using Palace, 2025. URL https://arxiv.org/abs/2511.09041.

Zaikun Zhang. PRIMA: Reference implementation for Powell's methods with modernization and amelioration. http://www.libprima.net, 2023. Software.